PROJECT REPORT

on

Speaker recognition from noisy speech using HMM

Submitted in fulfilment of the requirements for the degree of

Bachelor of Technology

By

P RAMYASRI PRIYA DARSHINI

(228867601022)

Under the Supervision of

Dr. Manish Kumar Bajpai

Associate Professor, Department of CSE

National Institute of Technology, Warangal

Department of Computer Science and Engineering

National Institute of Technology, Warangal

India

Declaration

I, P Ramyasri priya darshini, of the Department of Computer Science, National Institute of Technology Warangal, declare that this report titled “Speaker recognition from noisy speech using HMM” is the result of my independent work.

All the figures, tables, equations, algorithms, and data presented herein are either my original contributions or have been clearly cited and referenced as appropriate. I affirm that this work has not been submitted for the award of any other degree or diploma. I understand that plagiarism or misrepresentation of any kind is considered a serious academic offense and may result in disciplinary action.

Ramyasri priya darshini

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Abstract

This project presents a robust speaker recognition system using Mel-Frequency Cepstral Coefficients (MFCC) and Hidden Markov Models (HMM) with integrated noise handling. The system aims to accurately identify speakers even in noisy environments. Audio samples are collected and organized speaker-wise, followed by preprocessing steps such as normalization, noise addition at different Signal-to-Noise Ratio (SNR) levels, and denoising using spectral subtraction. MFCC features are then extracted to capture essential speech characteristics and used to train Gaussian HMM models for each speaker. During testing, unknown audio samples undergo the same processing steps, and the trained models compute likelihood scores to identify the speaker. The system is evaluated using accuracy, confusion matrix, and classification metrics under varying noise conditions. Experimental results show that the system achieves a high training accuracy of about 99%, while testing accuracy remains around 89% at 60 dB SNR and decreases to approximately 78.47% at 20 dB SNR. These results demonstrate that the system maintains good performance even in noisy environments, making it suitable for real-world speaker recognition applications.

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CHAPTER 1

INTRODUCTION

This project focuses on developing a robust speaker recognition system capable of identifying individuals based on their unique voice characteristics, even in the presence of background noise. Speaker recognition has gained significant importance in modern applications such as voice-based authentication, security systems, and smart assistants. However, the performance of such systems is often affected by environmental noise, which distorts speech signals and reduces recognition accuracy. To address this issue, the proposed system incorporates noise handling techniques along with advanced signal processing methods to ensure reliable performance under real-world conditions.

The system integrates key components such as denoising techniques, feature extraction using Mel-Frequency Cepstral Coefficients (MFCC), and probabilistic modeling using Hidden Markov Models (HMM). These techniques work together to enhance speech quality, extract meaningful features, and model the temporal patterns of speech effectively. Furthermore, the system is evaluated under different noise levels to analyze its robustness and consistency. By doing so, the project demonstrates the ability of the system to maintain accuracy even when speech signals are degraded, making it suitable for practical real-world applications.

1.1 Background

In practical scenarios, speech signals are often affected by noise due to environmental factors such as traffic sounds, machinery, background conversations, and recording device limitations. The presence of noise in audio signals became a major concern with the widespread use of communication systems, mobile devices, and voice-based technologies. As these applications expanded into real-world environments, it became evident that clean speech assumptions were unrealistic, leading to the need for noise analysis and speech enhancement techniques.

Noise in speech signals is typically evaluated using the Signal-to-Noise Ratio (SNR), which measures the relative strength of the speech signal compared to background noise. A higher SNR indicates clearer speech, while a lower SNR represents more noise interference. This metric is widely used to analyze system performance under varying noise conditions.

To address the impact of noise, various denoising techniques have been developed in the field of speech processing. One commonly used method is spectral subtraction, which operates in the frequency domain by estimating the noise spectrum and subtracting it from the noisy signal. This helps in reducing unwanted noise components while preserving important speech information. Other approaches include filtering techniques, statistical methods, and modern deep learning-based enhancement models. These denoising methods play a crucial role in improving the quality of speech signals and enabling more accurate feature extraction for speaker recognition systems.

1.2 Research Objective

The main objective of this project is to design and develop a robust speaker recognition system that can accurately identify speakers even in the presence of noise. In real-world scenarios, speech signals are often degraded due to environmental disturbances, making it challenging for conventional systems to maintain high accuracy. Therefore, the system aims to incorporate effective noise handling techniques to enhance speech quality before further processing. By doing so, the project focuses on improving the reliability and consistency of speaker recognition under varying noise conditions.

To achieve this, the system integrates denoising methods with feature extraction using Mel-Frequency Cepstral Coefficients (MFCC) and probabilistic modeling using Hidden Markov Models (HMM). The objective is not only to extract meaningful and discriminative speech features but also to model the temporal characteristics of speech effectively. Additionally, the system is evaluated across different Signal-to-Noise Ratio (SNR) levels to analyze its performance and robustness. Overall, the project aims to develop a system that maintains high accuracy, adapts to noisy environments, and provides a scalable solution for real-world speaker recognition applications.

1.3 Problem Definition

The primary problem addressed in this project is the degradation of speaker recognition performance in the presence of noise. In real-world environments, speech signals are often corrupted by various types of noise such as environmental sounds, background conversations, and recording distortions. These disturbances significantly affect the quality of the audio signal, making it difficult for speaker recognition systems to extract reliable and discriminative features. As a result, the system may fail to correctly identify speakers, leading to reduced accuracy and reliability.

Traditional speaker recognition systems are typically designed and trained using clean speech data, which limits their ability to perform effectively in noisy conditions. Noise not only alters the spectral characteristics of speech but also impacts the feature extraction process, particularly MFCC features, which are sensitive to distortions. This leads to incorrect classification and increased misrecognition rates. Therefore, the key challenge is to design a system that can effectively handle noisy inputs, enhance the speech signal through denoising techniques, and extract robust features for accurate speaker identification. The goal of this project is to overcome these limitations by integrating noise reduction methods with advanced feature extraction and modeling techniques, ensuring consistent and reliable performance under varying real-world conditions.

CHAPTER 2

LITERATURE SURVEY

[1] Title: Speaker Recognition based on Mel-Frequency Cepstral Coefficients and Vector Quantization

Mangesh Balpande proposed a contactless voice-based attendance system. The system used MFCC for feature extraction and Vector Quantization (VQ) for speaker modeling. It achieved 85–90% accuracy across 24 speakers, showing effectiveness over touch-based systems.

[2] Title: Enhancement in Speaker Recognition for Optimized Speech Features using GMM, SVM and 1-D CNN

Sumita Nainan proposed a text-independent speaker recognition system using optimized MFCC features. The study compared GMM, SVM, and 1-D CNN models with Fisher score-based feature selection. The 1-D CNN achieved the best performance with an accuracy of 94.77%.

[3] Title: Speaker Identification using Deep Learning Models for Enhanced voice Biometrics

Akhilesh proposed a speaker identification system using a hybrid CNN-RNN architecture to capture spectral and temporal features. The system utilized Librosa for extracting MFCC and chroma features for improved performance. It achieved an accuracy of 92.3%, making it suitable for real-time biometric authentication.

[4] Title: Enhancement of Speaker Identification System Based on Voice Detection Techniques using Machine Learning

Khine Zin Oo proposed a text-dependent speaker identification system for Myanmar speakers. The system used ZCR and STE for voice activity detection and MFCC with FFNN for classification. It achieved 89.25% accuracy, showing that silence removal improves performance.

[5] Title: Optimizing voice Biometrics verification in Banking with Machine Learning for Speaker Identification

Oyebode Oluwatobi Oyewale proposed a voice biometric system for banking applications. The system used adaptive filtering with a hybrid CNN-LSTM model to handle device and environmental variations. It achieved 97% accuracy, ensuring reliable and secure user authentication.

[6] Title: Speaker Identification for Low-End Devices: A Secure Voice Biometric Solution for Mobile Banking

Oyebode Oluwatobi Oyewale proposed a lightweight speaker identification system for mobile banking. The system used a dual-model approach with Whisper and ECAPA-TDNN for effective feature extraction. It achieved 99.59% accuracy and included liveness detection to prevent spoofing attacks.

[7] Title: An Ensemble Approach for Speaker Identification from Audio files in Noisy Environment

Syed Shahab Zarin proposed an ensemble speaker identification system. The system combined RNN and DNN (MLP) with VAD and a softmax classifier for noise robustness. It achieved 99.4% accuracy by effectively filtering noisy frames and enhancing speaker recognition.

[8] Title: A Speaker Identification-Verification Approach for noise-corrupted and improved Speech using Fusion feature and Convolution Neural Network

Rohun Nisa proposed a CNN-based speaker identification system using feature fusion. The system combined MFCC, NPF, and NPCC to capture robust speech characteristics. It achieved 96.1% accuracy, showing strong performance in noisy environments.

[9] Title: GMM-Based Speaker Verification System with Hardware MFCC in SoC Design

Tsung-Han Tsai and Chiao-Li Wang proposed a text-dependent speaker verification system with hardware-based MFCC extraction. The system combined SoC implementation with GMM and HMM models for efficient processing. It achieved 93.3% accuracy with low latency and highpower efficiency.

[10] Title: Speaker Identification using voice print Biometrics

Intesar Abd Ul-Salam Asa'ad proposed a speaker identification system using MFCC for feature extraction and SVM for classification. The system was implemented in a MATLAB environment for efficient processing. It achieved 100% accuracy, demonstrating high reliability for biometric authentication.

TABLE 1: SUMMARY OF EXISTING SPEAKER RECOGNITION METHODS

S.No	Year	Authors	Method and description	Model Used	Dataset	Results	Remarks
1	2021	Mangesh Balpande et al.	The system uses MFCC-based feature extraction with preprocessing steps to enable a contactless biometric attendance system by matching voice features with a stored database.	Vector Quantization (VQ)	A collection of voice samples from 24 speakers	The system achieved an average identification accuracy of 85-90%	The accuracy is sensitive to environmental noise and changes in a speaker's voice due to health factors like a cough, cold, or throat infection
2	2021	Sumita Nainan et al.	The system uses spectral subtraction and optimized MFCC with delta features and Fisher score selection to improve speaker recognition while reducing computational complexity.	Gaussian Mixture Model (GMM), Support Vector Machine (SVM), and 1-D Convolutional Neural Network (1-D CNN)	ViDTIMIT dataset	The 1-D CNN achieved the highest accuracy of 94.77%. The optimized SVM model achieved 94.51% accuracy	Including dynamic features improved recognition accuracy, while 1-D CNN performed best but requires larger datasets for optimal performance.
3	2023	Akhilesh et al.	The system performs preprocessing with noise reduction, silence removal, and normalization, followed by feature extraction using	hybrid CNN-RNN	A sizable dataset of labeled voice recordings	Achieved 92.3% accuracy	The system provides a robust real-time biometric authentication solution, though misclassification may occur for speakers with similar voice characteristics.

			MFCC, spectral roll-off, and chroma features.				
4	2023	Khine Zin Oo et al.	The system improves speaker identification using AVD (ZCR and STE) for noise and silence removal, followed by MFCC-based feature extraction.	Feedforward Neural Network (FFNN)	A custom dataset of 10 Myanmar speakers	The system achieved an identification accuracy of 89.25%	The study shows that AVD-based silence and noise removal improves speaker identification performance, with future work focusing on advanced features and noise reduction methods.
5	2024	Oyebode Oluwatobi Oyewale et al.	The study optimizes banking voice biometrics by comparing enhancement techniques and applying preprocessing with noise removal, frequency stabilization, and data augmentation for robustness.	hybrid CNN-LSTM architecture	A custom dataset of 480 voice samples from 12 participants	the highest identification accuracy of 97.1%	Adaptive filtering effectively handles non-stationary noise but may face real-time limitations on resource-constrained devices due to higher computational complexity.
6	2024	Oyebode Oluwatobi Oyewale et al.	The study proposes a lightweight speaker identification system for mobile banking using preprocessing, embedding extraction, and liveness detection for secure authentication.	Random Forest	A custom dataset of 50 university students	The system achieved 99.59% identification accuracy	The system enhances USSD banking security by preventing voice cloning and replay attacks while maintaining robustness across languages and noisy conditions.
7	2024	Syed Shahab Zarin et al.	The methodology improves noise robustness using VAD with MFCC and spectrogram features and applies zero padding to handle variable-length audio.	Recurrent Neural Network (RNN)	Kaggle speaker recognition dataset	The proposed system achieved a high test accuracy of 99.4%	The study shows VAD with softmax improves noise resilience, with MLP converging faster than RNNs, and MFCC outperforming spectrogram features.
8	2024	Rohun Nisa et al.	The study enhances speaker recognition in noisy environments by fusing MFCC, NPF, and NPCC features with preprocessing-based speech enhancement.	1D-Frame Level-Feature Fusion-CNN	ITU-T speech dataset	the system achieved a 96.1% identification accuracy	Incorporating pitch and phase features improves noise resilience over MFCC-only systems and remains effective with limited training data.
9	2024	Tsung-Han et al.	The system implements a text-dependent speaker verification approach using SoC-based hardware-software co-design, with	Gaussian Mixture Model (GMM)	TIMIT dataset	The system achieved a 93.3% accuracy rate	Hardware implementation of MFCC improves execution speed, while performance remains stable at moderate noise levels but drops

			hardware-accelerated MFCC extraction and software-based model classification for real-time performance.				significantly at 0 dB SNR due to severe interference.
10	2025	Intesar Abd Ul-Salam Asa'ad et al.	The system uses MFCC to extract acoustic features through signal processing and classifies them using a hyperplane-based model for speaker recognition.	Support Vector Machines (SVM)	A local database containing audio recordings from five individuals	Achieved 100% identification accuracy	The system is efficient and robust for biometric authentication under mild noise and pitch variations, but performance degrades in heavy noise, requiring larger datasets for better generalization.

CHAPTER 3

PROPOSED METHODOLOGY

The proposed speaker recognition system is designed to achieve robust performance under noisy conditions by integrating feature extraction, noise modeling, denoising, and probabilistic classification. The overall framework consists of training speaker-specific models using clean speech and evaluating them under varying noise levels.

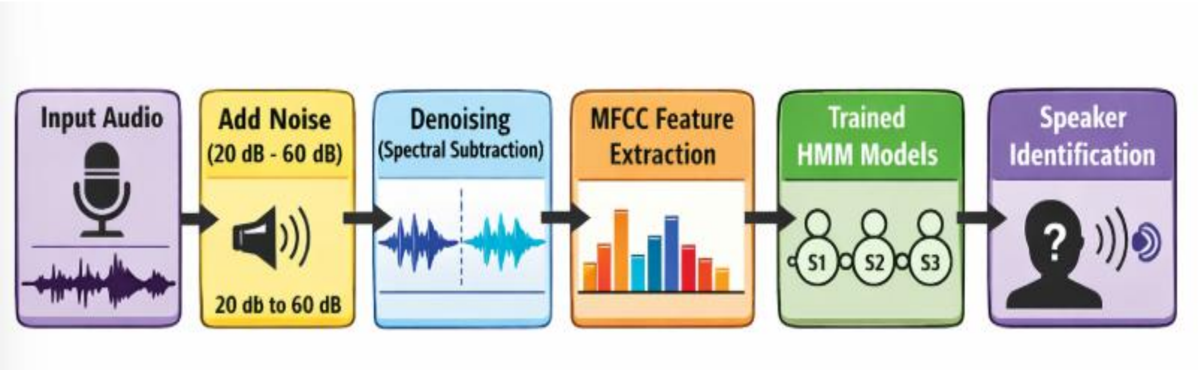

FIGURE 3.1: BLOCK DIAGRAM OF PROPOSED SYSTEM

3.1 Data Preparation

The speech dataset is organized in a speaker-wise manner, where each directory corresponds to a unique speaker. For each speaker, the available audio samples are randomly shuffled and partitioned into training and testing sets using an 80:20 split ratio. This ensures that the model is trained on a sufficient amount of data while maintaining unseen samples for evaluation.

3.2 Feature Extraction

Mel-Frequency Cepstral Coefficients (MFCCs) are extracted from each speech signal to capture the perceptually relevant spectral characteristics. Given an input speech signal, MFCC features are computed using a short-time analysis framework. To improve feature robustness and reduce variability due to recording conditions, Cepstral Mean and Variance Normalization (CMVN) is applied. The normalization is defined as:

$$\hat{x} = (x - \mu) / \sigma$$

where x denotes the original MFCC feature vector, μ represents the mean, and σ denotes the standard deviation. This normalization results in zero mean and unit variance, thereby improving the stability and robustness of the feature representation.

3.3 Noise Modeling

To simulate real-world noisy environments, White Gaussian Noise (WGN) is artificially added to the clean speech signals. The noise level is controlled using the Signal-to-Noise Ratio (SNR), defined as:

$$SNR = 10 \log_{10} (P\_signal / P\_noise)$$

where P_signal and P_noise represent the power of the speech signal and noise, respectively. In this work, multiple SNR levels ranging from 60 dB to 20 dB are considered to evaluate the robustness of the proposed system.

3.4 Speech Enhancement using Spectral Subtraction

To mitigate the effect of noise, a spectral subtraction-based denoising technique is applied to the noisy speech signals. The process involves transforming the signal into the frequency domain using Short-Time Fourier Transform (STFT), estimating the noise spectrum from initial frames, and subtracting it from the noisy magnitude spectrum. The enhanced magnitude spectrum is obtained as:

$$|S(f)| = \max (|Y(f)| - |N(f)|, 0)$$

where $|Y(f)|$ denotes the magnitude spectrum of the noisy signal and $|N(f)|$ represents the estimated noise spectrum. The denoised signal is then reconstructed using the inverse Short-Time Fourier Transform (STFT).

3.5 Speaker Modeling using Hidden Markov Model

Each speaker is modeled using a Gaussian Hidden Markov Model (HMM), which captures the temporal dynamics of speech. For a given observation sequence $O = (o1, o2, \ldots, oT)$, the likelihood of the model λ is given by:

$$P(O \mid \lambda)$$

The HMM parameters, including state transition probabilities, emission probabilities, and initial state distribution, are estimated using the Expectation-Maximization (EM) algorithm. In this work, each model is configured with 16 hidden states and diagonal covariance matrices.

3.6 Speaker Identification

During testing, the MFCC features of an input speech signal are extracted after noise addition and denoising. The likelihood of the observation sequence is computed for each speaker model. The predicted speaker is determined using the maximum likelihood criterion:

$$\hat{s} = \arg\max_s P(O \mid \lambda_s)$$

where λs represents the HMM corresponding to speakers. The speaker model yielding the highest likelihood score is selected as the identified speaker.

3.7 Performance Evaluation

The performance of the proposed system is evaluated using multiple metrics, including accuracy, confusion matrix, precision, recall, and F1-score. Additionally, robustness analysis is performed by plotting recognition accuracy against different SNR levels. This analysis provides insights into the system’s behavior under varying noise conditions and demonstrates the effectiveness of the denoising approach.

CHAPTER 4

RESULTS

4.1 Dataset Loading and Splitting

This output shows the successful loading and preprocessing of the dataset, along with the initialization of the model training process. Each speaker in the dataset is displayed along with the number of audio samples allocated for training and testing. The dataset is split using an 80:20 ratio, where the majority of samples are used for training and the remaining are used for testing.

```
PS D:\speaker_recognition> py train_and_test.py;
aew -> Train: 724 | Test: 182
ahw -> Train: 379 | Test: 95
aup -> Train: 379 | Test: 95
awb -> Train: 728 | Test: 182
axb -> Train: 379 | Test: 95
bdl -> Train: 724 | Test: 182
clb -> Train: 724 | Test: 182
eey -> Train: 380 | Test: 95
fem -> Train: 379 | Test: 95
gka -> Train: 379 | Test: 95
jmk -> Train: 727 | Test: 182
ksp -> Train: 724 | Test: 182
ljm -> Train: 379 | Test: 95
lnh -> Train: 724 | Test: 181
rms -> Train: 724 | Test: 182
rxr -> Train: 426 | Test: 107
slp -> Train: 379 | Test: 95
slt -> Train: 724 | Test: 182
```

FIGURE 4.1: LOADING DATASET AND SPLITTING

4.2 Training HMM Model:

Following the dataset distribution, the system begins training the Hidden Markov Models (HMM) for each speaker individually. For every speaker, a message indicating "Training model for <speaker>" is displayed, confirming that the model is being trained using the corresponding audio samples. This output verifies that the dataset has been properly organized and that the training process is being executed successfully for all speakers in the system.

```
Training model for aew        Training model for gka
Training model for ahw        Training model for jmk
Training model for aup        Training model for ksp
Training model for awb        Training model for ljm
Training model for axb        Training model for lnh
Training model for bdl        Training model for rms
Training model for clb        Training model for rxr
Training model for eey        Training model for slp
Training model for fem        Training model for slt
```

FIGURE 4.2: TRAINING HMM MODEL

4.3 Training Accuracy Evaluation:

This output presents the complete performance evaluation of the speaker recognition system. Initially, the system confirms that model training has been completed for all speakers and that the trained models have been successfully saved. The training accuracy is displayed as approximately **99.15%**, indicating that the model has effectively learned speaker-specific characteristics from the training dataset.

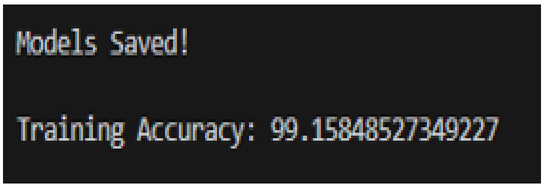

Figure 4.3:Training Accuracy

4.4 Performance Evaluation under Different Noise Levels:

The performance of the proposed speaker recognition system is evaluated under various noise conditions by testing with different Signal-to-Noise Ratio (SNR) levels ranging from 60 dB to 20 dB. At higher SNR levels such as 60 dB, the system achieves high accuracy (around 89%), indicating minimal impact of noise and effective speaker recognition. As the noise level increases (SNR decreases to 50 dB and 40 dB), a gradual reduction in accuracy is observed, showing the influence of noise on feature quality and model performance. At moderate noise levels like 30 dB, the accuracy further decreases to around 86%, reflecting increased distortion in the speech signal. At very low SNR levels such as 20 dB, the system experiences a significant drop in accuracy to approximately 78.47%, due to heavy noise interference which affects the clarity of speech features and increases misclassification. The confusion matrices and classification reports across these noise levels highlight that errors increase as noise intensifies. However, despite this degradation, the system maintains reasonable performance, demonstrating the effectiveness of spectral subtraction for denoising and the robustness of MFCC and HMM-based modeling in handling real-world noisy environments.

```
Testing with Noise + Spectral Subtraction (SNR=60 dB)
Testing Accuracy: 89.1773162939297

Confusion Matrix

[[160   0   0   0   0   5   0   0   0  14   3   0   0   0   0   0   0   0]
 [  0  88   0   0   0   0   0   0   0   0   7   0   0   0   0   0   0   0]
 [  0   0  84   0   0   0   1   0   0   1   5   0   0   1   0   0   3   0]
 [  0   0   0 180   0   0   0   0   0   0   2   0   0   0   0   0   0   0]
 [  0   0   0   0  91   0   0   0   0   0   0   0   0   0   0   0   4   0]
 [  1   0   0   0   0 174   0   0   1   5   0   0   0   1   0   0   0   0]
 [  0   0   0   0   0   0 175   5   0   0   0   0   1   0   0   0   1   0]
 [  0   0   0   0   0   0   0  92   0   0   0   0   1   0   0   0   2   0]
 [  0   0   0   1   0   1   0   0  91   0   2   0   0   0   0   0   0   0]
 [  0   0   0   0   0   0   0   0   0  92   2   0   0   0   0   1   0   0]
 [  1   0   0   0   0   0   0   0   0   2 178   0   0   0   1   0   0   0]
 [  0   0   0   3   0   1   0   0   1   4  10 157   0   1   3   2   0   0]
 [  0   0   0   0   0   0  16   7   0   2   0   0  61   0   0   0   9   0]
 [  1   0   0   0   0   3   2   3   0   1   0   0   1 149   0  20   1   0]
 [  0   0   0   3   0   1   0   0   0   5   6   1   0   0 165   1   0   0]
 [  0   1   0   0   0   0   0   0   0   1   4   0   0   0   0 101   0   0]
 [  0   0   0   0   0   0   0   2   0   0   0   0   1   0   0   0  92   0]
 [  0   0   0   0   1   0  64   7   0   0   2   0   0   0   0   0   5 103]]
```

FIGURE 4.4.1: PERFORMANCE EVALUATION AT 60DB SNR

```
Testing with Noise + Spectral Subtraction (SNR=50 dB)
Testing Accuracy: 89.05750798722045

Confusion Matrix

[[160   0   0   0   0   5   0   0   0  14   3   0   0   0   0   0   0   0]
 [  0  88   0   0   0   0   0   0   0   0   7   0   0   0   0   0   0   0]
 [  0   0  84   0   0   0   1   0   0   1   5   0   0   1   0   0   3   0]
 [  0   0   0 181   0   0   0   0   0   0   1   0   0   0   0   0   0   0]
 [  0   0   0   0  90   0   0   0   0   0   0   0   0   0   0   0   5   0]
 [  1   0   0   0   0 174   0   0   1   5   0   0   0   1   0   0   0   0]
 [  0   0   0   0   0   0 175   5   0   0   0   0   1   0   0   0   1   0]
 [  0   0   0   0   0   0   0  92   0   0   0   0   0   0   0   0   3   0]
 [  0   0   0   1   0   1   0   0  91   0   2   0   0   0   0   0   0   0]
 [  0   0   0   0   0   0   0   0   0  92   2   0   0   0   0   1   0   0]
 [  1   0   0   0   0   0   0   0   0   2 178   0   0   0   1   0   0   0]
 [  0   0   0   3   0   1   0   0   1   4  10 157   0   1   3   2   0   0]
 [  0   0   0   0   0   0  17   7   0   2   0   0  61   0   0   0   8   0]
 [  1   0   0   0   0   3   2   3   0   1   0   0   1 149   0  20   1   0]
 [  0   0   0   3   0   1   0   0   0   5   6   1   0   0 165   1   0   0]
 [  0   1   0   0   0   0   0   0   0   1   4   0   0   0   0 101   0   0]
 [  0   0   0   0   0   0   0   2   0   0   0   0   1   0   0   0  92   0]
 [  0   0   0   0   1   0  66   8   0   0   2   0   0   0   0   0   5 100]]
```

FIGURE 4.4.2: PERFORMANCE EVALUATION AT 50DB SNR

```
Testing with Noise + Spectral Subtraction (SNR=40 dB)
Testing Accuracy: 88.53833865814697

Confusion Matrix

[[159   0   0   0   0   5   0   0   0  15   3   0   0   0   0   0   0   0]
 [  0  88   0   0   0   0   0   0   0   0   7   0   0   0   0   0   0   0]
 [  0   0  84   0   0   0   1   0   0   1   5   0   0   1   0   0   3   0]
 [  0   0   0 181   0   0   0   0   0   0   1   0   0   0   0   0   0   0]
 [  0   0   0   0  89   0   0   0   0   0   0   0   0   0   0   0   6   0]
 [  1   0   0   0   0 174   0   0   1   5   0   0   0   1   0   0   0   0]
 [  0   0   0   0   0   0 174   5   0   0   0   0   2   0   0   0   1   0]
 [  0   0   0   0   0   0   1  91   0   0   0   0   0   0   0   0   3   0]
 [  0   0   0   1   0   1   0   0  91   0   2   0   0   0   0   0   0   0]
 [  0   0   0   0   0   1   0   0   0  91   2   0   0   0   0   1   0   0]
 [  1   0   0   0   0   0   0   0   0   2 178   0   0   0   1   0   0   0]
 [  1   0   0   3   0   1   0   0   1   5  12 154   0   0   3   2   0   0]
 [  0   0   0   0   0   0  15   7   0   4   0   0  60   0   0   0   9   0]
 [  1   0   0   0   0   3   2   3   0   1   0   0   1 149   0  20   1   0]
 [  0   0   0   4   0   1   0   0   0   5   6   1   0   0 164   1   0   0]
 [  0   1   0   0   0   0   0   0   0   1   5   0   0   0   0 100   0   0]
 [  0   0   0   0   0   0   0   2   0   0   0   0   1   0   0   0  92   0]
 [  0   0   0   0   1   0  68   8   0   0   2   0   0   0   0   0   5  98]]
```

FIGURE 4.4.3: PERFORMANCE EVALUATION AT 40DB SNR

```
Testing with Noise + Spectral Subtraction (SNR=30 dB)
Testing Accuracy: 86.14217252396166

Confusion Matrix

[[151   0   0   0   0   3   0   0   0  24   4   0   0   0   0   0   0   0]
 [  0  89   0   0   0   0   0   0   0   0   6   0   0   0   0   0   0   0]
 [  0   0  83   0   0   0   1   0   0   1   7   0   0   1   0   0   2   0]
 [  0   0   0 180   0   0   0   0   0   0   2   0   0   0   0   0   0   0]
 [  0   0   0   0  85   0   0   0   0   0   0   0   0   0   0   0  10   0]
 [  1   0   0   0   0 173   0   0   1   7   0   0   0   0   0   0   0   0]
 [  0   0   0   0   0   0 173   6   0   0   0   0   1   1   0   0   1   0]
 [  0   0   0   0   0   0   2  89   0   0   0   0   0   0   0   0   4   0]
 [  0   0   0   1   0   1   0   0  93   0   0   0   0   0   0   0   0   0]
 [  0   0   0   0   0   0   0   0   0  92   1   0   0   0   0   2   0   0]
 [  1   0   0   1   0   0   0   0   0   1 178   0   0   0   1   0   0   0]
 [  0   3   0   2   0   0   0   0   2   6  17 145   0   0   5   2   0   0]
 [  1   0   0   0   0   0  18   6   0   8   0   0  51   0   0   0  11   0]
 [  0   0   0   0   0   4   4   1   0   2   0   0   2 148   0  19   1   0]
 [  0   0   0   6   0   1   0   0   0   7   5   1   0   0 161   1   0   0]
 [  0   1   0   0   0   0   0   0   0   4   5   0   0   0   0  97   0   0]
 [  0   0   0   0   0   0   0   1   0   1   0   0   1   0   0   0  92   0]
 [  0   0   0   0   2   0  84   7   0   1   1   0   4   0   0   0   6  77]]
```

FIGURE 4.4.4: PERFORMANCE EVALUATION AT 30DB SNR

```
Testing with Noise + Spectral Subtraction (SNR=20 dB)
Testing Accuracy: 78.47444089456869

Confusion Matrix

[[112   0   0   0   0  20   0   0   0  42   8   0   0   0   0   0   0   0]
 [  0  91   0   0   0   0   0   0   0   0   3   0   0   0   0   1   0   0]
 [  0   0  71   1   0   2   1   0   0   8   8   0   0   1   0   0   3   0]
 [  1   0   0 172   0   0   0   0   0   1   8   0   0   0   0   0   0   0]
 [  0   0   0   0  63   0   1   3   0   1   0   0   3   0   0   0  24   0]
 [  1   0   0   3   0 171   0   0   1   5   0   0   0   1   0   0   0   0]
 [  2   1   0   0   0   0 114  32   0   1   2   0   5   1   0   0   8  16]
 [  0   0   0   0   0   0   5  79   0   5   0   0   2   0   0   0   3   1]
 [  0   0   0   6   0   1   0   0  85   0   3   0   0   0   0   0   0   0]
 [  0   0   0   0   0   0   0   0   0  83   9   0   0   0   0   3   0   0]
 [  0   0   0   1   0   0   0   0   0   1 177   0   0   0   2   1   0   0]
 [  1   6   0  16   0   1   0   0   2  13  28 103   0   0  11   1   0   0]
 [  1   0   1   0   0   4  11   1   0   9   0   0  57   4   0   0   7   0]
 [  0   0   0   0   0  10   4   1   0   7   0   0   2 141   0  14   2   0]
 [  0   0   0  10   0   3   0   0   0  11  13   1   0   0 143   1   0   0]
 [  0   6   0   0   0   0   0   0   0   2  17   0   0   0   0  82   0   0]
 [  0   0   0   0   0   0   0   1   0   5   1   0   0   0   0   0  88   0]
 [  0   0   0   0   1   0  23   9   0   1   3   0   8   0   0   0   4 133]]
```

FIGURE 4.4.5: PERFORMANCE EVALUATION AT 20DB SNR

4.5 System Performance at Varying SNR Levels:

The graph illustrates how your speaker recognition system performs as the environment becomes increasingly noisy, specifically measuring **Accuracy** against the **Signal-to-Noise Ratio (SNR)**. At the highest SNR of **60 dB**, which represents nearly crystal-clear audio, your Hidden Markov Model (HMM) reaches its peak performance of approximately **89% accuracy**. As you introduce more background noise and move left toward **20 dB**, the accuracy drops to about **78%**. This decline happens because noise distorts the acoustic features—like the spectral transitions—that the HMM relies on to move between states and identify the speaker's unique "voice print." Interestingly, the curve shows a sharp improvement between **20 and 40 dB** before leveling off, suggesting that while your model is quite robust in moderate noise, the high-interference environment at 20 dB is where the HMM's ability to statistically "guess" the correct speaker from corrupted audio sequences is most heavily tested.

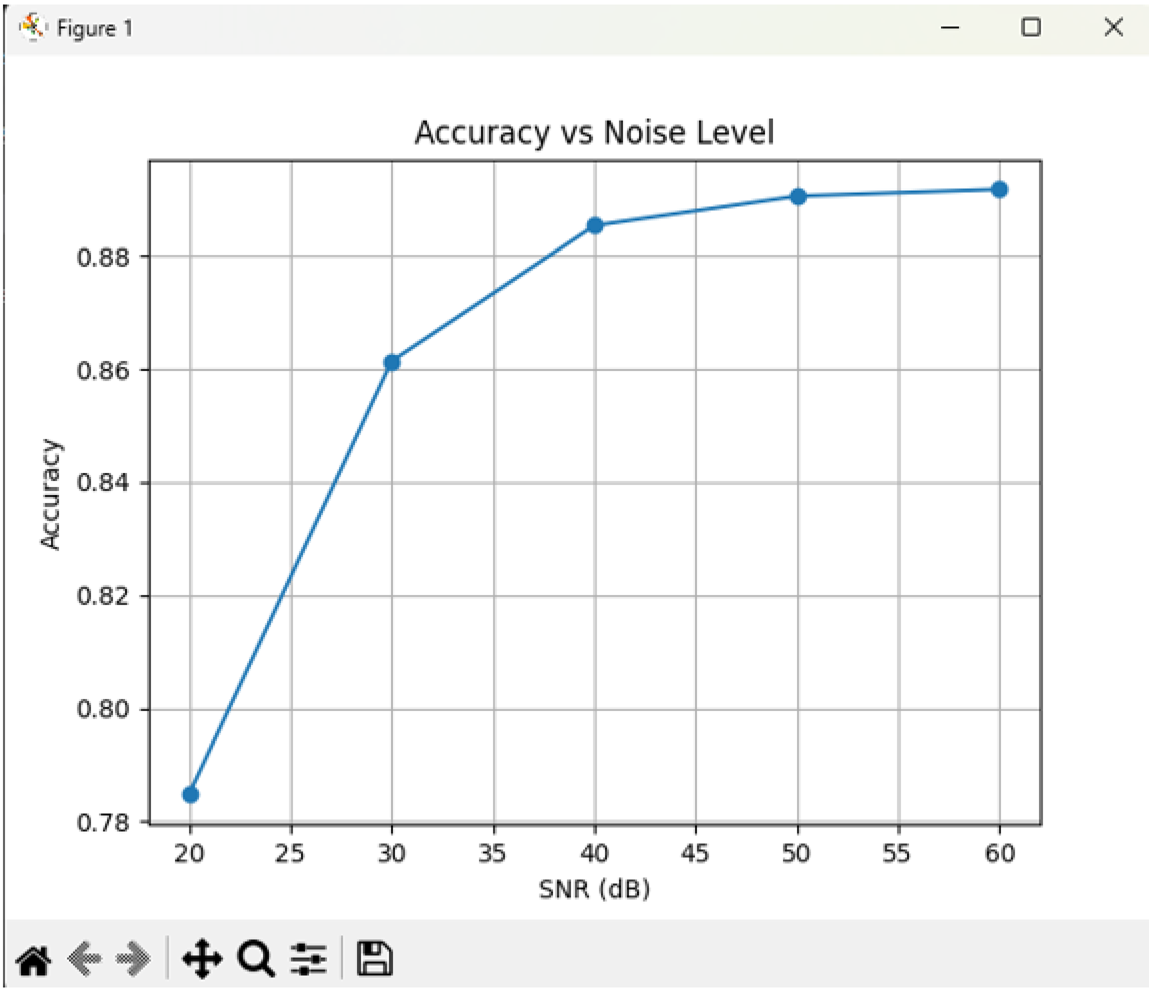

FIGURE 4.5: SYSTEM PERFORMANCE AT VARYING SNR LEVELS

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

5.1 Conclusion

The proposed speaker recognition and denoising system using MFCC and Hidden Markov Models (HMM) demonstrates an effective approach for identifying speakers under both clean and noisy conditions. The system successfully integrates noise addition, spectral subtraction-based denoising, feature extraction using MFCC, and probabilistic modeling using HMM to achieve reliable performance.

The experimental results show that the system achieves high training accuracy of around 99%, indicating that the model effectively learns speaker-specific characteristics. During testing, the system maintains strong performance at higher Signal-to-Noise Ratio (SNR) levels, with accuracy around 89% at 60 dB. As noise increases, a gradual decrease in accuracy is observed, reaching approximately 78.47% at 20 dB. This trend highlights the impact of noise on speech signals while also demonstrating the robustness of the proposed system.

The use of spectral subtraction significantly improves speech quality, enabling better feature extraction and classification even in noisy environments. Additionally, MFCC features effectively capture the perceptual aspects of speech, and HMM provides a powerful framework for modeling temporal variations in voice signals.

Overall, the system proves to be efficient, reliable, and suitable for real-world applications such as voice-based authentication and secure access systems. Despite performance degradation at very low SNR levels, the system maintains acceptable accuracy, confirming its robustness and practical applicability.

5.2 Future Scope

1. **Integration with Advanced Deep Learning Models:** The system can be enhanced by incorporating deep learning architectures such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, or Transformer-based models. These models can capture more complex patterns in speech, potentially improving recognition accuracy, especially in highly noisy or reverberant environments.

2. **Support for Multi-Language and Dialect Recognition:** Expanding the system to handle multiple languages, accents, and dialects would make it more versatile for global applications. This would involve training on a larger, more diverse dataset.

3. **Real-Time Processing Capabilities:** Implementing the system in a real-time framework would enable live speaker verification for applications such as secure access control, voice-based personal assistants, and surveillance systems.

4. **Advanced Noise Robustness:** Future improvements could include integrating more sophisticated noise reduction algorithms, adaptive filtering, and data augmentation techniques to handle extremely noisy environments or overlapping speech scenarios.

5. **Deployment on Mobile and Embedded Platforms:** Optimizing the system for resource-constrained devices would allow for portable speaker recognition solutions in smartphones, IoT devices, and wearable technology.

6. **Integration with Multi-Modal Biometric Systems:** Combining voice recognition with other biometric methods such as facial recognition or fingerprint scanning could provide enhanced security and reliability in sensitive applications.

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